



Where the **agentic stack** is being built.

Agents Talking To Agents

MCP, A2A, and the Reality of Multi-Agent Orchestration in Production

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Why I'm Talking About This

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AI & Agentic Security

Building AI-driven security workflows and multi-agent systems for real operational environments.

AI Orchestration in Production

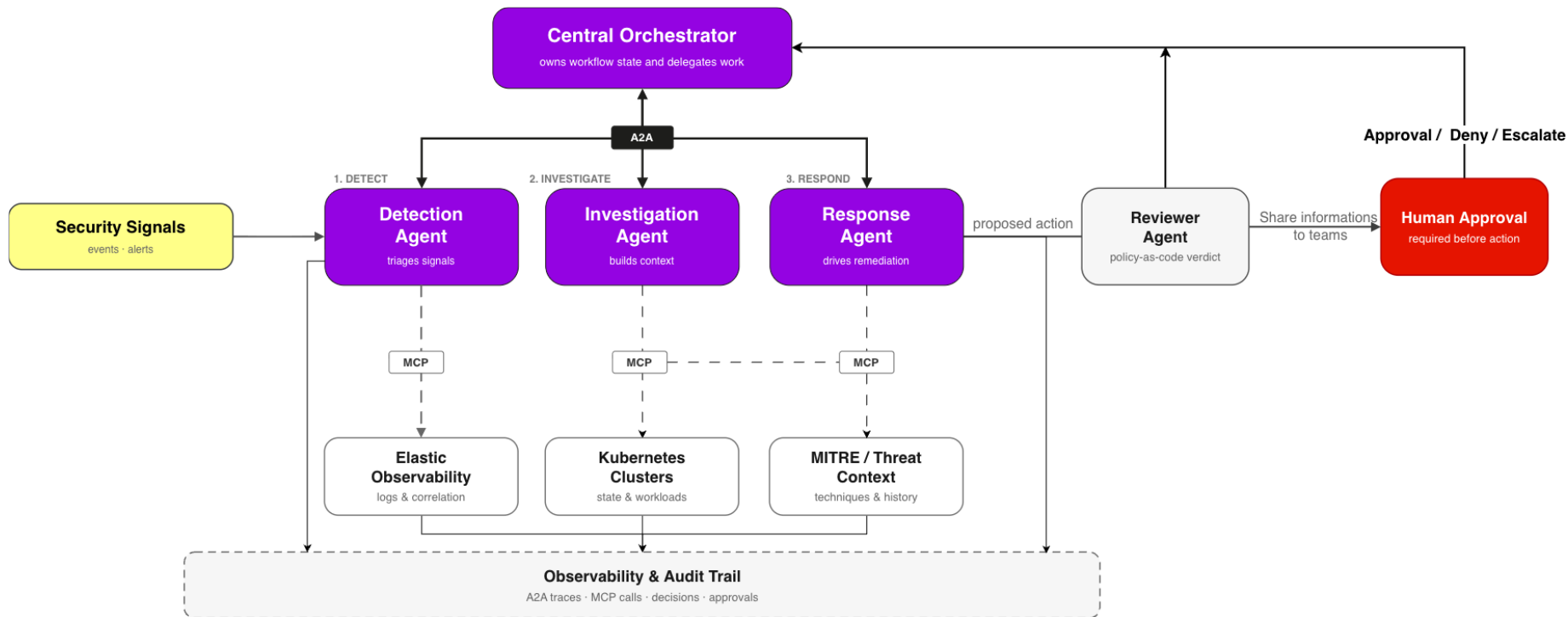
Designing agent orchestration, tool use, decision flows and automation around security use cases.

Security at Network Scale

Applying AI to 5G Core, fixed and home networks, Telco Cloud and Cloud-Native environments.



The Obvious Architecture



Multi-Agent Architecture: What We Assume



Context



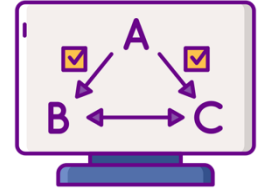
Ownership



Authority



Retry



Causality

The next agent
has the full
picture.

Delegating the
task transfers
responsibility.

Delegation
limits what the
agent can do.

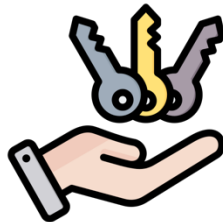
A retry
repeats one
failed
operation.

Green protocol
calls mean
workflow
success.

Multi-Agent Architecture: What Actually Breaks



Context



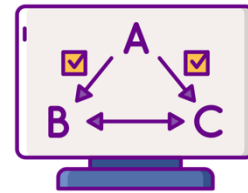
Ownership



Authority



Retry



Causality

The critical constraint is missing, stale, or distorted.

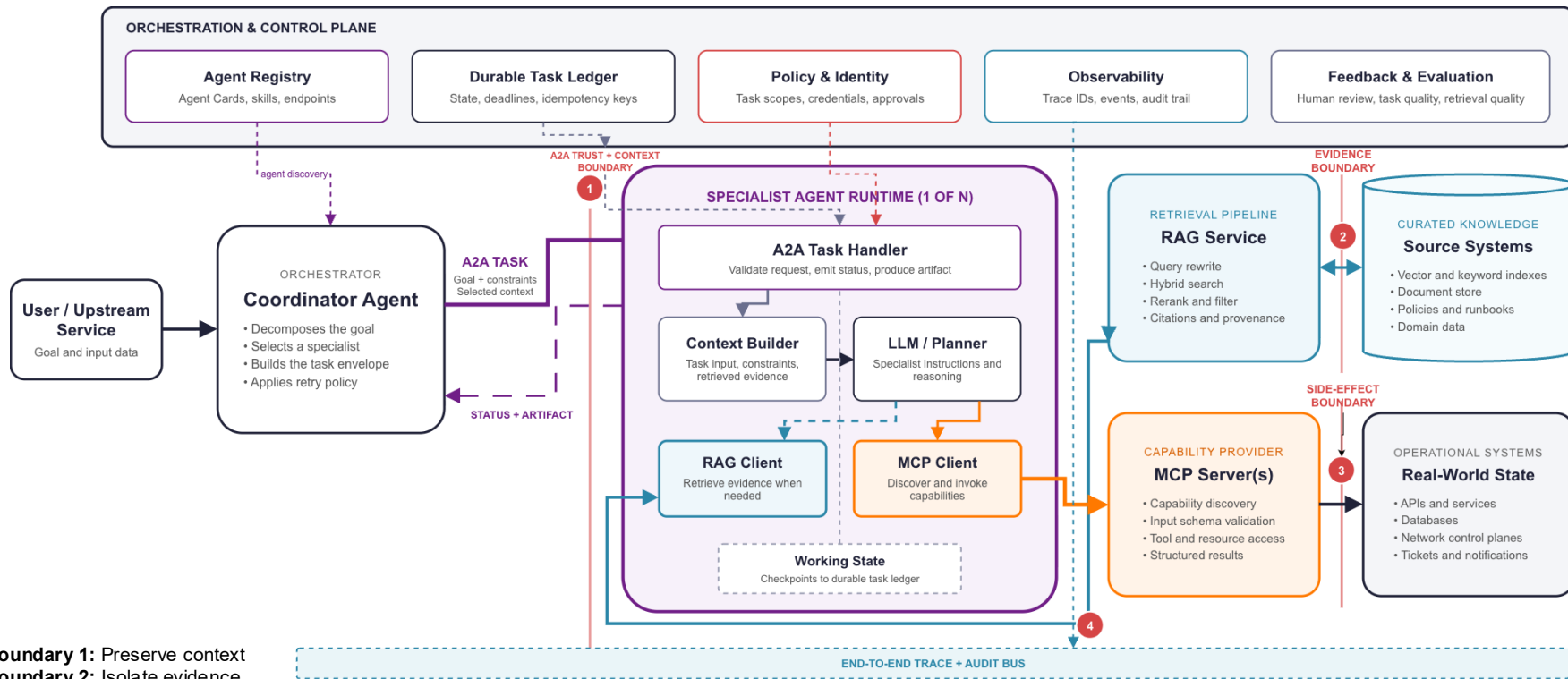
Execution moves.
Validation and closure have no explicit owner.

Permissions may exceed the task scope.

The outcome is unknown.
The next attempt may duplicate or diverge.

Each step succeeds locally. The final outcome is wrong and unexplained.

Pattern 1: A2A Delegation, MCP Execution

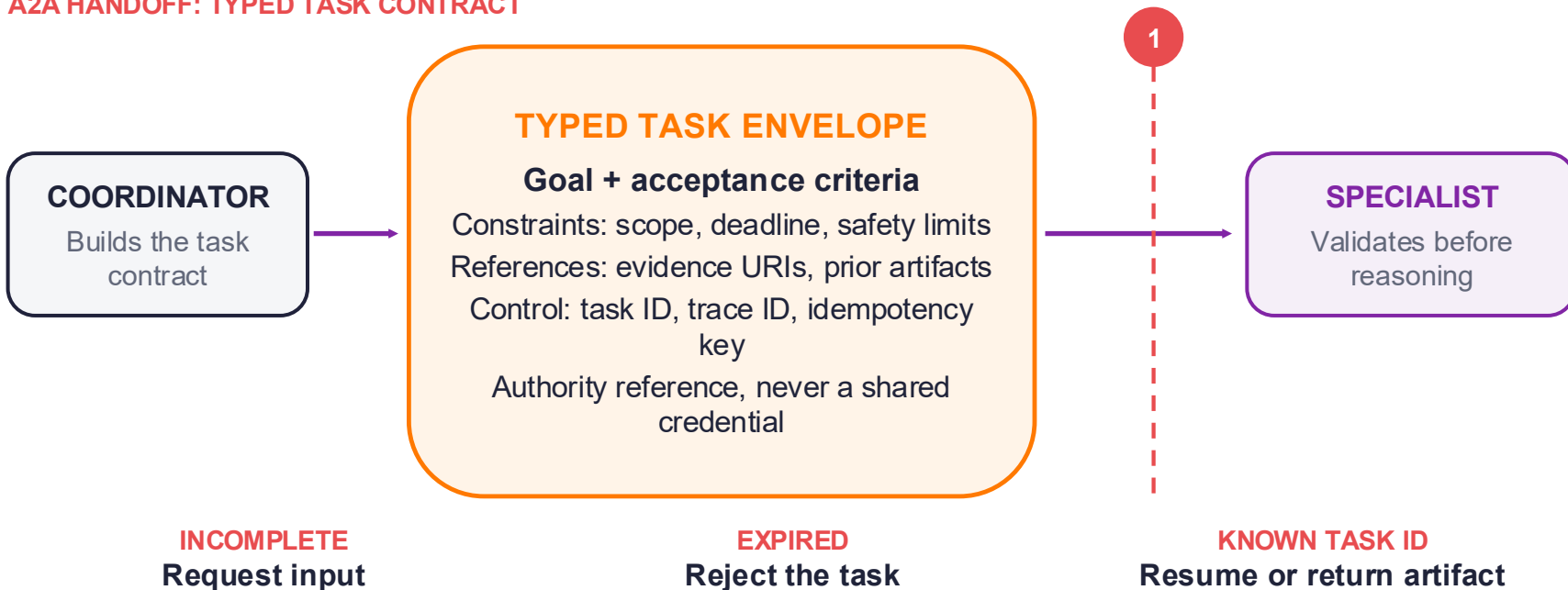


Boundary 1: Preserve context
Boundary 2: Isolate evidence
Boundary 3: Control mutations
Boundary 4: Verify outcomes

Where the agentic stack is being built.

Boundary 1: Context Transfer

A2A HANDOFF: TYPED TASK CONTRACT



Validate the contract before the receiving agent reasons or acts.

Boundary 1: Context, Cache, and Durable State

WHY BOUNDARY 1 NEEDS A LEDGER

PROMPT CONTEXT

Instructions and evidence assembled for the current run

ONE MODEL RUN

KV CACHE

Cached attention state for shared prefixes

SERVING RUNTIME

DURABLE STATE

Task status, artifacts, policy decisions, and observed outcomes

WORKFLOW LIFETIME

Caching saves compute. The ledger preserves ownership, recovery state, and audit evidence.

The ledger carries the workflow across model runs.

Operational Foundation: Reproducible Agent Runs

VERSION THE INPUTS, RECORD THE EFFECTS

■ MODEL + INFERENCE CONFIG

■ SYSTEM PROMPT + TASK
TEMPLATE

■ TASK INPUT + EVIDENCE IDS →

■ TOOL SCHEMAS + POLICY
VERSION

RUN MANIFEST

task_id · trace_id
input hashes + version set
tool calls + artifact references
approvals + observed outcome

SAFE REPLAY

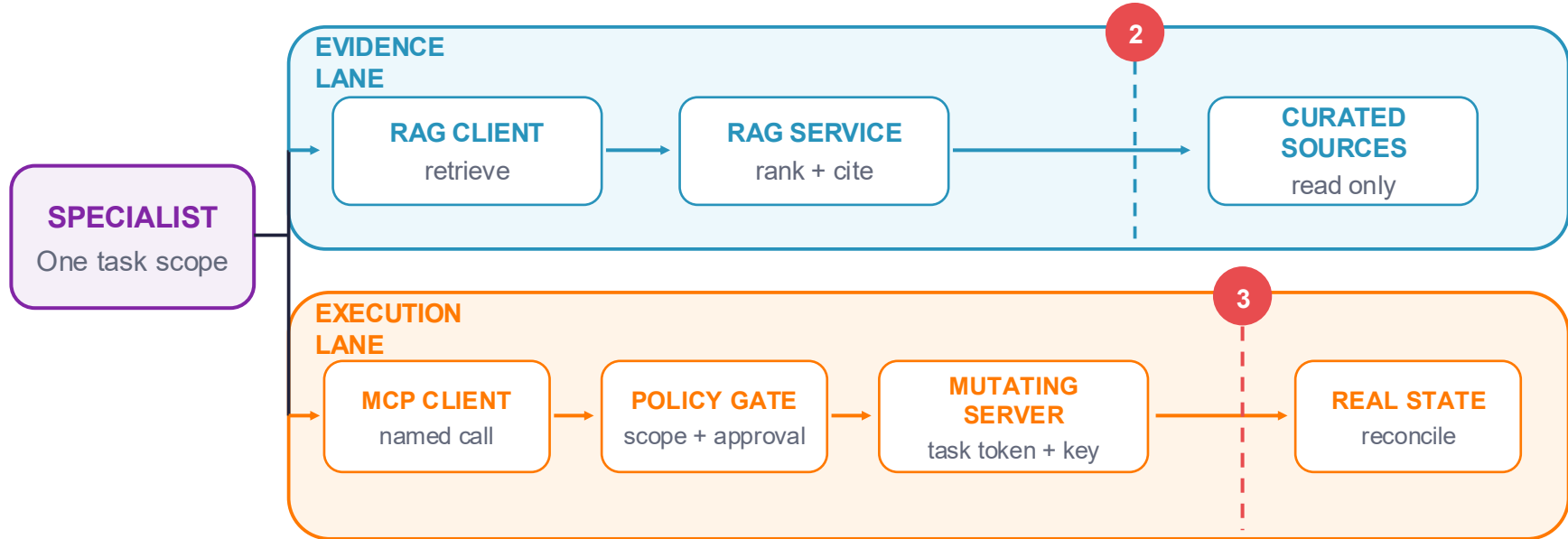
- 1 Rebuild recorded inputs
- 2 Compare decisions + outcome
- 3 Revalidate before mutation

Token sequences may vary even when the execution contract is preserved.

Reliable replay preserves the conditions and revalidates every side effect.

Pattern 2: Task-Scoped Authority

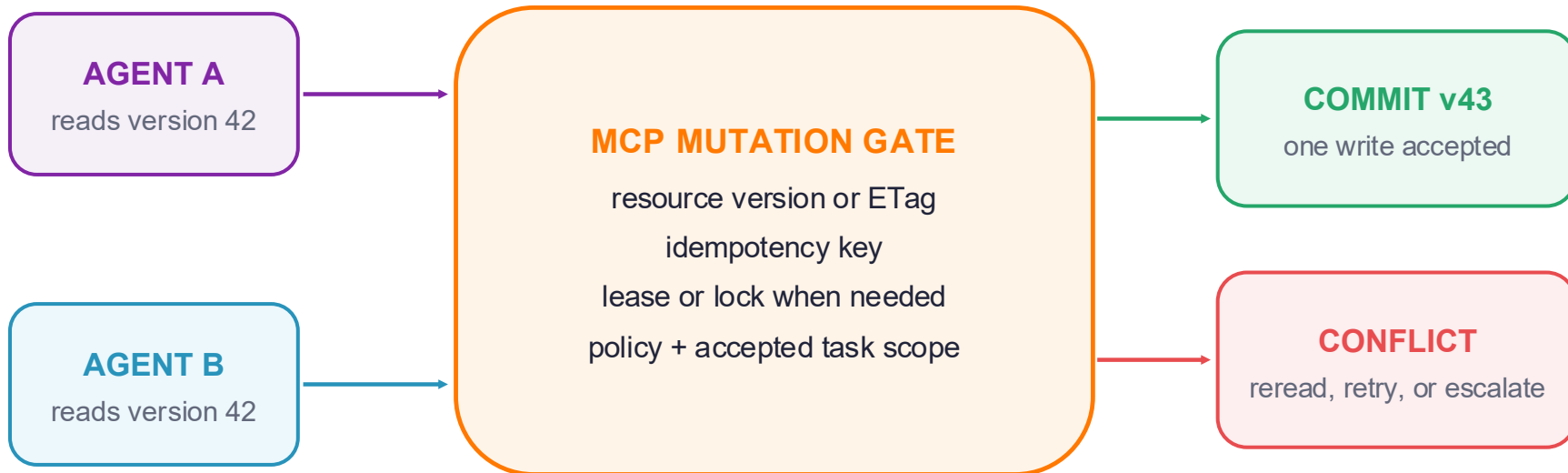
BOUNDARIES 2 + 3: READ EVIDENCE, CONTROL SIDE EFFECTS



Task scope comes from the handoff. Policy enforces it at the MCP boundary.

Boundaries 2 & 3: Concurrent Agents, One Resource

TOOL CONFLICTS SURFACE UNDER REAL LOAD



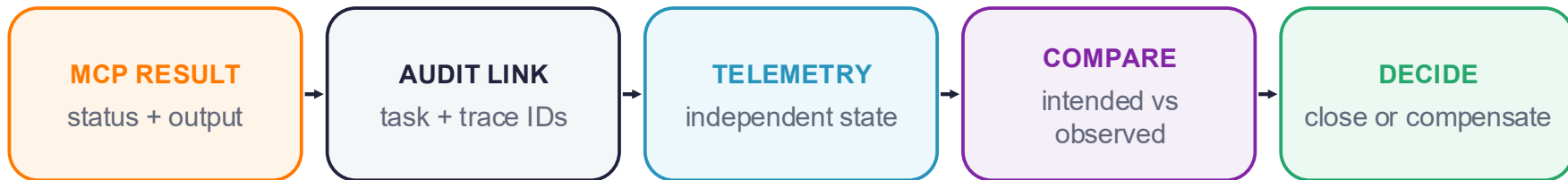
Reconciliation reads the resource after the write and records the observed state.

Authority answers who may act. Concurrency control decides which write commits.

Boundary 4: Outcome Verification

CAUSALITY: PROTOCOL STATUS + OBSERVED STATE

4



FAILURE MODE

A successful call can target the wrong resource or produce no operational effect.

task_id trace_id artifact_id resource_id observed_state

Close the task only after telemetry and orchestrator confirm the intended state.

Pattern 3: Recoverable A2A Handoffs

BOUNDARIES 1 + 4: RECOVERY AFTER A PARTIAL FAILURE



Verified checkpoints let the next actor resume, compensate, or escalate.

Three Production Patterns, Four Failure Boundaries

APPLY THESE AT THE SYSTEM BOUNDARIES

1

SEPARATE ROLES

Agent + capability boundaries

A2A delegates work. MCP exposes bounded capabilities.

2

SCOPED AUTHORITY

Boundaries 2 + 3

Separate retrieval from mutation. Issue authority per task.

3

RECOVERABLE HANDOFFS

Boundaries 1 + 4

Checkpoint artifacts. Resume, compensate, or escalate.

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What We Would Change Today

PRODUCTION RETROSPECTIVE

- 01 ORCHESTRATED STATE** A durable workflow engine owns transitions; LLMs reason inside bounded specialist nodes.
- 02 TASK-SCOPED EXECUTION** Policy issues short-lived authority. The MCP server verifies scope before mutation.
- 03 GOVERNED FEEDBACK** Verified outcomes drive evaluation; humans review ambiguity and high-impact change.

Approval everywhere creates reviewer fatigue and weaker oversight.
Risk tiers direct human attention to critical edge cases.

The end !



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Agentic AI Foundation

AGNTCon + MCPCon Europe

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